



Patrick Dwyer

Data Scientist with 3 years of experience in cloud-based data engineering and front-end development. Proficient in designing, developing, and maintaining scalable systems across AWS, Azure, Fabric, and GCP. Strong expertise in Python, SQL, Javascript, CSS, HTML, and Terraform, with hands-on experience building data pipelines, machine learning models, containerized applications, CI/CD workflows, interactive web applications & reports, and more. A fast learner with a problem-solving mindset, passionate about practical applications of technology. Google Cloud [Certified](#) Professional Machine Learning Engineer.

Experience

Manifold Group: *Data Scientist*

October 2023 – Present

- Productionized NLP research code, designing and architecting a scalable solution that reduced runtime from 6 hours to 30 minutes (~92% improvement), enabling faster iteration and broader production use.
- Implemented vector embedding similarity pipelines from inception to production, unlocking the ability for clients to analyze previously opaque free-form text and derive actionable insights.
- Optimized client stored procedures decreasing query time by 99.8% to enable ingestion of > 100GBs of data
- Developed and maintained data pipelines to ingest and process > 1TB of data from a client's database into a data lakehouse in Microsoft Fabric
- Led, designed, and implemented enterprise-level data solutions for a Fortune 500 client using PySpark and Microsoft Fabric
- Worked in a team setting to develop terraform resources and design networking for a data lakehouse architecture on Azure
- Designed and developed anomaly detection application (Python, Docker, AWS EC2)

Websanity: *Web Developer*

September 2023 – Present

- Freelance Web Developer for [Rulepop](#), an emerging rules reference platform for tabletop games
- Implemented and maintaining Javascript features on an as needed basis

Schwartz Lab: *Lab Tech*

July 2023 – Sep 2023

- Developed supervised 3d Convolutional Neural Network in PyTorch
- Manually labelled 8,544 ground truth 2d points to ensure model accuracy

Projects

[cwdmdc.patrickdwyer.com](#)

Winter 2025-Present

- Responsive CWD sampling results dashboard built with MDC's public ArcGIS API

[patrickdwyer.com](#)

Summer 2023-Present

- Responsive CV website built with industry best practices using vite & vanilla html/css/js
- Integrated offline PWA functionality using [vite-pwa-plugin](#)

Education

BA Mathematics, BA Computer Science

Sep 2019 – June 2023

Northwestern University, Evanston, IL — Weinberg College of Arts and Sciences — 3.43/4.00 GPA